Non-Equilibrium Green's Functions (NEGF) and their application to Quantum Conductance in nano-junctions

The lectures are meant for postgraduate and good 4th year undergraduate students, although they could also be useful to anybody wishing to familiarise her/himself with NEGF and their applications to the phenomenon of quantum conductance. The idea is to introduce NEGF generally, then consider the problem of quantum conductance in nano-junctions in tight-binding model (where the problem can be solved exactly) and then, time permitting, discuss possible generalisations including the GW approximation. The lectures will be mostly mathematical, the intention being to help in reading literature on the subject; there will be very little time for applications of the technique, although some attempts to illustrate the taught material will be made when appropriate.

A tentative plan of lectures is given below.

Introductory lectures (6 lectures)

Lecture 1. Second quantisation technique (Slater determinants; Slater rules; second quantisation for electrons; Hamiltonian, electron density and current operators).

Lecture 2. Time evolution operators for general time-dependent Hamiltonians in quantum statistical mechanics (density matrix, general solution of the Liouville equation, quantum statistical averages of operators, Keldysh contour)

Lecture 3. Various Green's functions (definitions; G, G-lesser, G-greater, retarded, advanced; properties, including boundary conditions; averages of operators via G; energy, density, current; interaction representation and the Keldysh contour; examples for the tight-binding (TB) Hamiltonian)

Lecture 4. Equation of motion method (equation of motion for G; hierarchy of equations; definition of self-energy; Hartree-Fock approximation; Dyson equation; Keldysh equations)

Lecture 5. Time-ordered integrals (Langreth rules; Kadanoff-Baym equations; examples of applications)

Lecture 6. Quantum conductance (a general junction and its TB Hamiltonian; leads "self-energy"; famous expression for the current; transmission probability; scf with respect to the electron density; TranSiesta; some illustrations)

This plan is only tentative as it is difficult at this stage to predict whether the time allocated for each of the lectures (2h) is sufficient to cover the topic or indeed is too long for that. Appropriate corrections will be taken in due course and hence each lecture may take less or more than the 2h slot. There might be hand-written notes, but at the moment I cannot guarantee this.

Additional lectures

I do not expect these additional material to be given during the planned two weeks, although some of it might be possible to fit in. However, if the introductory lectures prove to be useful, then I will consider a second set of lectures to cover this bit some time before the end of the year.

Lecture 7. Expansions of self-energy (functional derivatives; self-energy via functional derivative; GW; Hedin's diagrams; density response)

Lecture 8. General discussion of going beyond the TB Hamiltonian in considering quantum conductance (e.g. exclusion of TB leads; additivity of the self-energy; Feynman diagrams for G; GW for the central region)

Lecture 9. Second quantisation for phonons. Quantum heat conductance.

Lecture 10. Matsubara Green's functions (equation of motion; properties, boundary conditions; Fourier series; Lehmann representation and physical meaning; calculation)

[May be something else?]